Name: Date: Period:

Chapters 20-21 HW (Circuits)

This HW is due on Mini-Test Day, **Fri, May 8**.

Questions and figures are from OpenStax *College Physics*, 2e (Chapter 20-21) unless otherwise noted.

Ohm's Law Problems (Chapter 20)

1. (P2) A total of 600 C of charge passes through a flashlight in 0.500 hr. What is the average current?
2. (P18) What current flows through the bulb of a 3.00-V flashlight when its hot resistance is $3.60\,\Omega$?
3. (P21) How many volts are supplied to operate an indicator light on a DVD player that has a resistance of $140\,\Omega$, given that $25.0\,\mathrm{mA}$ passes through it?

Circuits Concepts

4. (CQ2) Car batteries are rated in ampere-hours (A h). To what physical quantity do ampere-hours correspond (voltage, charge, current, resistance, etc), and what relationship do ampere-hours have to energy content?
5. (Original Question) Is current directly proportional or inversely proportional to voltage? Why does this make sense?

Name: Date: Period:

Equivalent Resistance Problems (Chapter 21)

6. (P2) (a) What is the resistance of a 100-Ω, a 2.50-kΩ , and a 4.00-kΩ resistor connected in series? (b) In parallel?

7. (P3) What are the largest and smallest resistances you can obtain by connecting a 36.0-Ω, a 50.0-Ω, and a 700-Ω resistor together?

8. (Original Problem) Find the current going through the battery in the circuit shown in Figure 1.

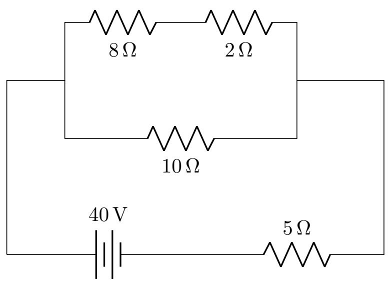

Figure 1: Original Figure.

9. (Giancoli Ch 19 P16) Determine (a) the equivalent resistance of the circuit shown in Figure 2, (b) thee voltage acros each resistor, and (c) the current through each resistor.

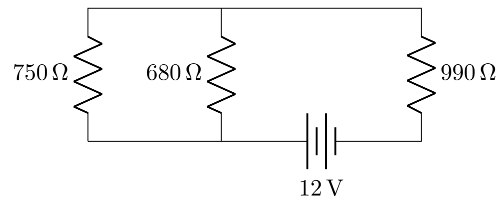

Figure 2: Giancolli Figure 19-48.

Name: Date: Period:

Equivalent Resistance Concepts

10. (Original Question) What happens to resistance when bulbs are wired in series? Why does this make sense?

11. (Original Question) What happens to resistance when bulbs are wired in parallel? Why does this make sense?

Additional Practice

12. (P1) (a) What is the resistance of ten 275-Ω resistors connected in series? (b) In parallel?

13. (Original Problem) Find the current going through the battery in the circuit shown in Figure 3.

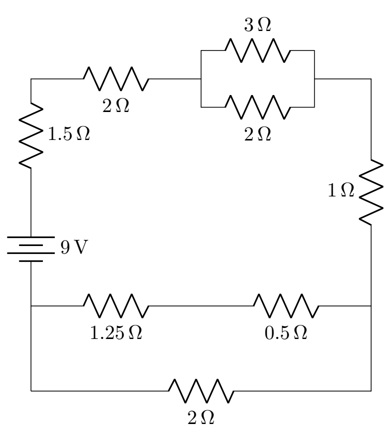

Figure 3: Original Figure.

Bonus

14. Calculate the equivalent resistance of a cube that has a 1-ohm resistor along each edge, as shown in Figure 4,

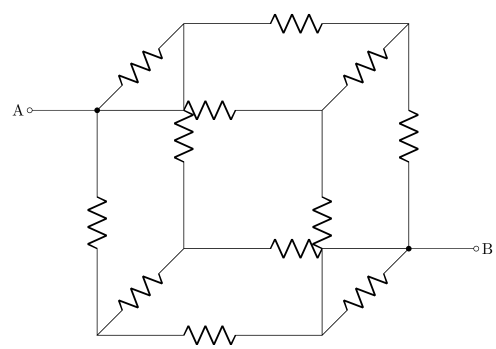

Figure 4: Original Figure. Each resistor has a value of 1 Ω

Equations

Electrostatics Equations

$$F = \frac{kq_1q_2}{r^2} \qquad E \equiv \frac{F}{q} = \frac{kq}{r^2} \qquad V \equiv \frac{PE}{q} = -\frac{W}{q}$$

Circuits Equations

$$I \equiv \frac{\Delta q}{t} \qquad V = IR \qquad P = IV = I^2R = \frac{V^2}{R}$$

$$R_{eq} = R_1 + R_2 + \cdots \qquad \frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \vdots$$

Unit Conversions

$$1\,\text{mC} = 10^{-3}\,\text{C} \qquad 1\,\mu\text{C} = 10^{-6}\,\text{C} \qquad 1\,\text{nC} = 10^{-9}\,\text{C}$$

Constants

Coulomb Constant	$k = 9.0 \times 10^9\,\text{Nm}^2/\text{C}^2$
Charge of electron/proton	$e = \pm 1.60 \times 10^{-19}\,\text{C}$
Electron mass	$m_e = 9.11 \times 10^{-31}\,\text{kg}$
Proton mass	$m_p = 1.673 \times 10^{-27}\,\text{kg}$
Neutron mass	$m_n = 1.675 \times 10^{-27}\,\text{kg}$

Selected Answers

1. 333 mA
2. 0.833 A
3. 3.50 V
6. (a) $6.60\,\text{k}\Omega$; (b) $\frac{20}{213}\text{k}\Omega = 0.094\,\text{k}\Omega$
7. $786\,\Omega$; $20.3\,\Omega$
8. 4.00 A
9. Equivalent: $R = 1346.6\,\Omega$, $I = 8.9$ mA;
   990 Ω: $V = 8.8$ V, $I = 8.9$ mA
   680 Ω: $V = 3.2$ V; $I = 4.3$ mA
   750 Ω: $V = 3.2$ V; $I = 4.7$ mA
12. (a) $2750\,\Omega$; (b) $27.5\,\Omega$
13. 1.36 A